UbiPower18000_16x15.36TB_NVMe_4xNDR400_3xStorageNode_16hosts256GBMem

System Description

System Name: UbiPower18000_16x15.36TB_NVMe_4xNDR400_3xStorageNode_16hosts256GBMem

Submitter Name: UBIX TECHNOLOGY CO., LTD

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Benchmark Version: MLPerf Storage v2.0

Solution Type: Proprietary - UBIXFS(distributed filesystem)

Submission Division: CLOSED

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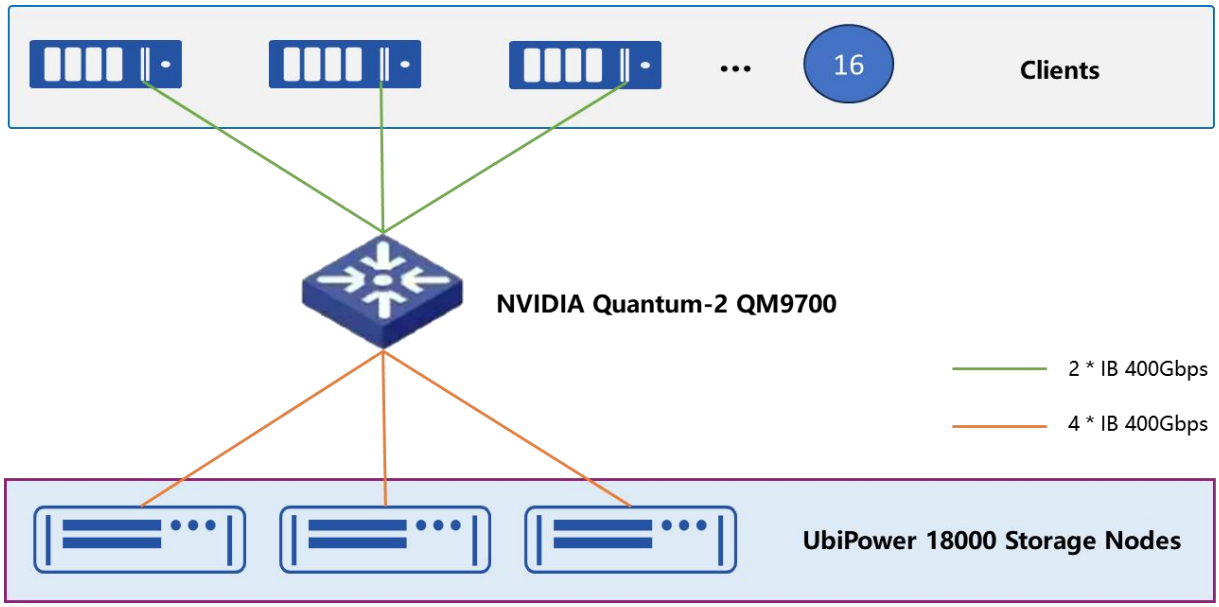
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1. Hardware

The hardware is composed of 16 clients and 3 servers connected to the same switch.



1.1. Client

The clients is a commercial x86 architecture server node with 2 CPU sockets and 256GB of RAM. All clients are configured the same (hw & sw).The clients communicate with the server through IB using the two links dedicated for storage.

The detailed configuration of the client is shown in the following table：

Client Node components	Configuration Description	Client Node Count
Client CPU	Intel(R) Xeon(R) Platinum 8468 (*2)	16
Client Memory	256GB DDR5 (64GB *4)	
Client Network Type	Mellanox CX7 Single-port IB NDR 400Gb (*2)	

1.2. Server

The server is a commercial UbiPower 18000 distributed all-flash storage, based on the x86 server architecture,engineered for high-performance with low latency and high scalability.
The detailed configuration of the servers is shown in the following table：

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Server Node components	Configuration Description	Server Node Count
Server CPU	INTEL(R) XEON(R) GOLD 5520+ (*2)	3
Server Memory	1024GB DDR5 (32GB *32)	
Server Network Type	Mellanox CX7 Single-port NDR IB 400Gb (*4)	
Data drives	DapuStor R6100 15.36TB NVMe SSD (*16)	

1.3. Switch

Mellanox(NVIDIA) Quantum-2 QM9700 switch,all clients and servers are connected to it.

1.4. Power & Rack Requirements

For detailed system power requirements, see the table below：

System component	Power supply unit	Nameplate Rated Power(W)	Design Power(W)	Rack Unit
Storage Node 1	Power supply 1	1300	1300	2
	Power supply 2	1300		
Storage Node 2	Power supply 1	1300	1300	2
	Power supply 2	1300		
Storage Node 3	Power supply 1	1300	1300	2
	Power supply 2	1300		
Totals		7800	3900	6

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2. Software

2.1. Client

The detailed configuration of the client-side software is shown in the table below：

Software Type	Software Configuration/Version
Client OS	Ubuntu 22.04.5 LTS
Client Kernel	5.15.0-119-generic
UBIXFS	2.1.0
Mellanox OFED	MLNX_OFED_LINUX-24.10-0.7.0.0

2.2. Server

The detailed configuration of the server-side software is shown in the table below：

Software Type	Software Configuration/Version
Server OS	UbiPower OS
Server Kernel	2.1.0
Storage	UbiPower 18000
Mellanox OFED	MLNX_OFED_LINUX-23.10-3.2.2.0

2.3. Switch

NA

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3. Settings

3.1. Client

1. Install standard UBIXFS POSIX client on the host with default settings.
2. The file system is mounted on the client using UBIXFS.
3. Limit the client's memory to 256GB via GRUB configuration.

3.2. Server

For detailed configuration information about storage servers, see the table below：

Storage server Configuration	**Description**
Vendor Name	UBIX TECHNOLOGY CO., LTD
Product Model	UbiPower 18000
Version	2.1.0
Redundancy Policy	EC 4+2:1

3.3. Switch

NA

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4. Configuration

4.1. Storage config

--rdma=true

4.2. Client config

--rdma=true --direct_io=true

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